

Modbus Variable Speed Drive [VSD] MAPS Template (Electrical)

The Modbus Variable Speed Drive [VSD] MAPS template is designed to control a motor that is connected to a Mitsubishi VSD on Modbus.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is a Modbus VSD, which provides control over the motor connected to the Mitsubishi VSD on Modbus.

This document describes the Basic Modbus Variable Speed Drive (VSD) Motor MAPS template. For details of its required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Modbus VSD Functional Philosophy

The VSD start command output is enabled once all interlocks are ready and a start request is received.

Field Interlocks

- Trip Feedback
- Fault Feedback

Interlock fault recovery process – once a field interlock is activated the faceplate will display a red indication, if the interlock is recovered and still latched in the PLC function block and the faceplate will show a green indication and the reset button on the faceplate will flash, pressing the Reset button will unlatch the interlock and display a grey indication and allow the Modbus VSD to become ready for operation.

Start and stop requests are initiated by selecting a mode on the Modbus VSD faceplate:

- **Auto mode** - the auto control program initiates a start and stop, for normal PLC control.
- **Manual mode** - the operator can start and stop the motor from the faceplate.
- **Hand mode** - the motor is in PLC bypass mode.

Typical Faceplate Graphics for the Modbus VSD Motor MAPS Template

DOL ISO	DOL Aerator	DOL Standard Pump Side View Left	DOL PUMP LEFT	DOL PUMP RIGHT
	Hand Mode			
	Manual Mode			
	Auto Mode			
	Motor Control Name			
Graphics legend:				
Line colour	Fill colour	DOL status		
Black	Grey	Stopped		
Grey	White	Stopping		
Green	White	Starting		
Black	Green	Running		
Black	Yellow/Red	Fault Active		

Signal Description		Agent Type	Basic
SCADA Control Words:			
SSCL 0	Control Word	Marshal	x
SSCL 1	Speed Set point	Analog	x

Signal Description		Agent Type	Basic
SCADA Status Words:			
SSSL 0	Status Word	Marshal	x
SSSL 1	Speed Feedback	Analog	x
SSSL 2	Current Feedback	Analog	x
SSSL 3	Total Running Hours	Analog	x

Digital Inputs:		Basic
DI 0	Hand (On = Hand)	x
DI 1	Auto (On = Auto)	x
DI 2	Trip (On = Tripped)	x
DI 3	Fault (Off = Fault)	x

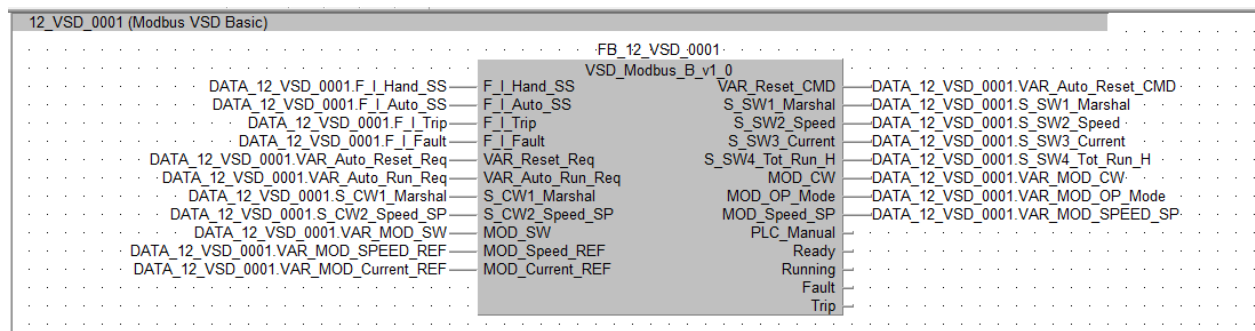
SCADA Control Word (Adroit Marshal Agent)		Event Logged	Basic
Bit 0	Alarms All Reset	x	x
Bit 1	Manual	x	x
Bit 2	Start Command	x	x
Bit 3	Stop Command	x	x

Bit 4	Spare		
Bit 5	Spare		
Bit 6	Spare		
Bit 7	Spare		
Bit 8	Spare		
Bit 9	Spare		
Bit 10	Spare		
Bit 11	Spare		
Bit 12	Spare		
Bit 13	Spare		
Bit 14	Spare		
Bit 15	Spare		

SCADA Status Word (Adroit Marshal Agent)		Alarmed	Basic
Bit 0	Hand		x
Bit 1	Auto		x
Bit 2	Manual		x
Bit 3	Ready		x
Bit 4	Starting		x
Bit 5	Running		x
Bit 6	Trip	x	x
Bit 7	Fault	x	x
Bit 8	Spare		
Bit 9	Spare		
Bit 10	Spare		
Bit 11	Spare		
Bit 12	Spare		
Bit 13	Spare		
Bit 14	Spare		
Bit 15	Spare		

Resources	Basic
Digital Inputs	4
Digital Outputs	0
Analog Inputs	0
Analog Outputs	0
PLC Memory Steps	296
System Words Used	23
System Bits Used	118
SCADA SCAN Points	6

Template: Modbus VSD PLC Function Block



Template Graphic: Home Screen	Control	Descriptions
	Manual	Manual control by the operator
	On	Manually start the DOL
	Off	Manually stop the DOL
	Reset	Reset the DOL after a fault has been repaired
	Healthy	Healthy to start the DOL

Template Graphic: Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.